



**EMN DEFENCE
SANAYİ ve DIŞ TİCARET A.Ş.**

Hürriyet Bulvarı No:129 / 4 Beylikdüzü/ İstanbul / Türkiye
Tel.: 0212 999 05 69 e-mal: info@emn defence.com



HAKKIMIZDA

EMN DEFENCE Silah Sanayi A.Ş. Türkiye ve dünyanın her yerindeki müşterilerine silah ve yedek parça tedarik ve üretiminde, talepler doğrultusunda yüksek rekabet gücü ile hizmet vermektedir.

EMN DEFENCE Silah Sanayii A.Ş., dünyanın dört bir yanındaki kanalları aracılığıyla müşterilerine son teknoloji silah ve silah parçalarının üretimini sağlamaktadır.

EMN DEFENCE silah sanayi A.Ş. müşterilerimizin askeri ve savunma ihtiyaçlarını karşılamak ve en uygun çözüm ve seçenekleri sunmak için dünyanın en büyük ve en iyi üretim zincirine sahiptir.

ABOUT US

EMN DEFENCE Silah Sanayii A.Ş. serves its customers in Turkey and all over the world with high competitiveness in the supply and production of weapons and spare parts in line with the demands.

EMN DEFENCE Silah Sanayii A.Ş. provides its customers with the production of state-of-the-art weapons and weapon parts through its channels all over the world.

EMN DEFENCE has the largest and best production chain in the world to meet the military and defense needs of our customers and to offer the most suitable solutions and options.

VİZYONUMUZ

Firmamız halihazırda ihracat yapan ve savunma sanayi alanında ihracat yapmayı hedefleyen bir firmadır.

Yatırıma karar verilmeden önce sektör analizi yapıldı, potansiyel pazarlar ve müşteriler belirlendi.

Yatırım kapsamındaki ürünler tamamen yerli ve milli kaynaklarla

Türk mühendis ve teknisyenler tarafından geliştirilecek.

Ürünlerin üretim bilgi paketleri tarafımızca geliştirilecek ve savunma sanayinde önemli bir tasarım kültürü oluşturulacaktır.

OUR VISION

Our company is a company that already exports and aims to export in the field of defense industry.

Before deciding on the investment, the sector was analyzed and potential markets and customers were identified.

The products within the scope of the investment will be developed entirely by Turkish engineers and technicians with domestic and national resources.

The production information packages of the products will be developed by us and an important design culture will be created in the defense industry.

MİSYONUMUZ

Yıllar içinde yaptığımız çalışmalar sonucunda geliştirdiğimiz know-how Her şeyden önce ülkemize hizmet etmek görevimizdir.

Silah ve mühimmat tedariki ile başlayacak olan hizmetimizin, aktif olarak sürdürdüğümüz Ar-Ge faaliyetleri sonucunda milli savunma sanayisini en üst düzeye çıkarmasını temenni ediyoruz.

OUR MISSION

The know-how we have developed as a result of our work over the years is our duty to serve our country above all else.

We hope that our service, which will start with the supply of weapons and ammunition, will bring the national defense industry to the highest level as a result of our active R&D activities.

KASKLAR / HELMETS



ESS1100



ESS1101



ESS1102



ESS1103



ESS1104



ESS1105



ESS1106



ESS1107

ŞAPKALAR / HATS



ESS1201



ESS1202



ESS1203



ESS1204



ESS1205



ESS1206





ESS1301



ESS1302



ESS1303



ESS1304



ESS1305



ESS1306



ESS1307



ESS1308



ESS1309



ESS1310



ESS1311



ESS1312



ESS1313



ESS1314



ESS1315



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ESS1320





ESS1401



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ESS1414



ESS1415



ESS1416



ESS1417



ESS1418



ESS1419



ESS1420



BOTLAR / BOOTS



ESS1501



ESS1502



ESS1503



ESS1504



ESS1505



ESS1506



ESS1507



ESS1508



ESS1509



ESS1510



ESS1511



ESS1512



ESS1513



ESS1514



ESS1515

ELDİVENLER / GLOVES



ESS1601



ESS1602



ESS1603



ESS1604



ESS1605



ESS1606



ESS1607



ESS1608



ESS1609



ESS1610



ESS1611



ESS1612



ESS1613



ESS1614



ESS1615



POLİS KIYAFETLERİ / POLICE CLOTHES



POLİS KALKANLARI / POLICE SHIELDS





ESS1901



ESS1902



ESS1903



ESS1904



ESS1905



ESS1906



ESS1907



ESS2001



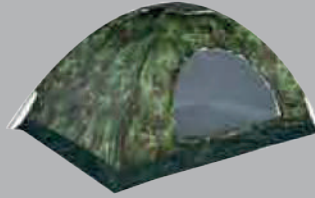
ESS2002



ESS2003



ESS2004



ESS2005



ESS2006



ESS2007



ESS2008



ESS2009



ESS2010

GIDA PAKETLERİ / FOOD PACKAGES



ESS2101



ESS2102



ESS2103



ESS7801



ESS7802



ESS7803



ESS7804



ESS7805



ESS7806



ESS7807



ESS7808





ESS5001

PART	PROPERTIES
Caliber	9 x 19 mm
Total Length With Stock Retracted	30.2 inch
Total Length (Stock Extended)	33.7 inch
Height	9 inch
Weight (Without Magazine)	2400 gr
Barrel Length	15 inch
Barrel Groove Set Number	1:10 / 6 pcs right-loop groove-set
Muzzle Velocity	396 m/s
Effective Range	100 m
Barrel Life (Not Forged Barrel)	10.000 rounds minimum
Muzzle Break	A2 Type or A2 interface with 3 prongs
Working Principal	Blow Back
Type Of Fire	Safety, Semi Automatic, or (+Full Automatic)
Sight Range	15.8 inch
Magazine Type	STANAG box type composite or aluminum
Magazine Capacity	15 Rd
Trigger Pull Force	2,50 - 3 kg / 25 - 30 N
Handguard Type	Lower and upper pikatiny, right and left M-LOK
Handguard Lenght	12 inch
Covering	Type III Hard Coat Anodized
Production Standart	C.I.P. / SAAMI
Warranty	2 Year



ESS5002



ESS5003



ESS5004



ESS5005



ESS5006



ESS5007



ESS5008

PART	PROPERTIES
Brand	MKA
Model	556
Caliber	5.56 x 45 mm (.223 Rem)
Total Length With Stock Retracted	870mm for 18,11" Barrel
Total Length (Stock Extended)	960 mm for 18,11" Barrel
Height	230 mm
Weight (Without Magazine)	32500gr for 18,11" Barrel
Barrel Length	10,3" beetwen 22"
Barrel Groove Set Number	6 pcs right-loop groove-set
Barrel Life (Not Forged Barrel)	10.000 rounds minimum
Muzzle Break	A2 Type or A2 interface with 3 prongs
Working Principal	Direct impingement
Type of Fire	Safety, Semi Automatic, or (+Full Automatic)
Sight Range	300 mm
Magazine Type	STANAG box type composite or aluminum
Magazine Capacity	30 Rd
Trigger Pull Force	25 - 45 N
Rate of Fire	750 -950 rp
Muzzle Velocity	880 m/s
Effective Range	500 m
Covering	Anodized
Production Standart	C.I.P. / SAAMI
Warranty	2 Year





ESS5009



ESS5010



ESS5011

MFR 56

MFR56, designed by DEFENCE, is produced in Defense factories in Turkey.

The MFR56 (MULTIFUNCTIONAL RIFLE) 5.56×45mm Caliber is a multifunctional machine gun. It can be produced as 2 different systems as OPEN BOLT and CLOSED BOLT due to the preferences of country-based law enforcement agencies.

Manufactured by DEFENCE, the MFR56 is especially designed for local operations. In areas and campuses that restrict the movement and maneuvering capabilities of security units, the MFR56 transforms the disadvantages of the field into an advantage for its user with its dimensions, weight and functions. MFR56 Design and development processes have been completely completed with the tests and feedback of SPECIAL FORCES field units, and it has been made ready for use with realistic operation simulations. The MFR56 is designed as a dual feed. There are 4 different patents belonging to DEFENCE in the upper feeding system.

ESS5012



AR-15

System Defence Upper wiht Barrel

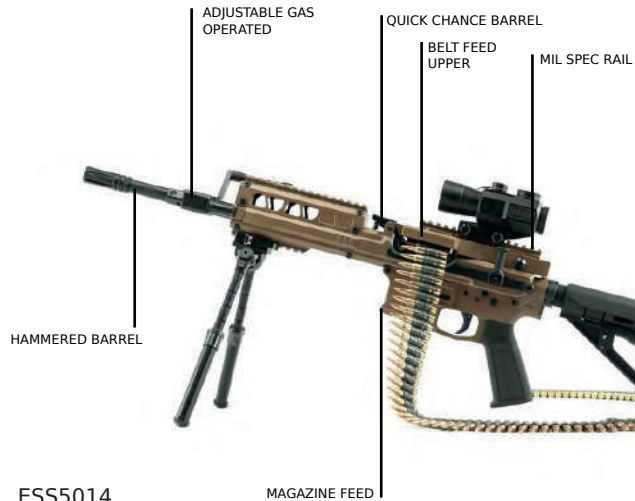
Belt-fed upper receiver assembly is designed to interchange with standard AR/M4

DUAL FEED 5.56X45 NATO



ESS5013

HIGHLIGHTS MFR56



ESS5014

9x18 mm



ESS2201

9x18 mm



ESS2202

9 mm



ESS2203

M04



ESS2204

M99



ESS2205

M99-40x46mm



ESS2206

7.62 mm



ESS2207

14.5x114 mm



ESS2208

12.7x108 mm



ESS2209

7.62 mm



ESS2210

7.62 mm



ESS2211

14.5x114mm



ESS2212

5.45 mm



ESS2213

7.62 mm



ESS2214

7.62x54 mm



ESS2215

12.7x108 mm



ESS2216

Cooled Thermal Binocular
NTG5000CZ

ESS10001



Military Thermal Binocular
NTG5000

ESS10002



Handheld Thermal Binocular
TB675

ESS10003



Fusion Thermal Monocular
Rattlesnake-800IRC

ESS10004



Clip On Thermal Imager
PAS29C

ESS10005



Sniper Detector Laser Illuminator
Night Vision Binocular

ESS10006

V366L



Night Vision Monocular
NVM2229

ESS10007



Night Vision Goggle
YSG

ESS10008



Holographic Thermal Sight
THS32050

ESS10009



Night Vision Riflescope
PJ055-3 Series

ESS10010



DÜRBÜNLER / BINOCULARS

8X32

8X42



FMC

Fully Multi-Coated Lenses deliver maximum brightness and contrast



FOG-PROOF

Product is vacuumd and purged with pure Nitrogen to avoid fogging when using in low temperature



BIRDING

Product is designed with high optical performance that fit for birding



LANDSCAPING

Product fit for viewing scenic landscape

ESS100011

7X50 C



FLOATING

Product is built light-weight enough to be float in water



FOG-PROOF

Product is vacuumd and purged with pure Nitrogen to avoid temperature



LANDSCAPING

Product fit for viewing scenic landscape



NAVIGATION

Built-in compass and rangefinding reticle for finding your bearings while sailing on the ocean or hiking in the field.

ESS100012



MILITARY & LAW ENFORCEMENT

Product fit for using by military and/or law enforcement department



ALL-WEATHER

Product is constructed for use in different weather conditions



ACQUA-VIEWING

Product fit for viewing aquatic wildlife in open water

ESS100013



ZOOM

5x Optical zoom



HUNTING

Product fit for hunting wild games



DAYTIME COLOR

Can expect to get full color images at daytime



RECORDING

Photo shooting, video recording & Play back



SURVEILLANCE

Product can be used for navigation, surveillance and targeting



BOATING

Night Vision Binoculars for Marine & Boating uses



OUTDOOR

The Night Vision will allow you to enjoy night nature tours in outdoor

ESS100014



ZOOM

5x Optical zoom



HUNTING

Product fit for hunting wild games



DAYTIME COLOR

Can expect to get full color images at daytime



RECORDING

Photo shooting, video recording & Play back



SURVEILLANCE

Product can be used for navigation, surveillance and targeting



BOATING

Night Vision Binoculars for Marine & Boating uses



OUTDOOR

The Night Vision will allow you to enjoy night nature tours in outdoor

ESS100015

FREEVUE EYE BOX

Generous eye-relief of over 4 inch get you on target fast and easy



TAC-LOK TURRETS

Integrated tactical locking turrets provides fast, accurate, and readable windage and elevation adjustments that prevent accidental dialing

SIDE-FOCUS & IR CONTROL

FOCUS from 10m to infinity distance at side wheel. Turn On/Off and change reticle illumination brightness and color with press button



BFMC COATING

Broadband Fully Multi-Coated lenses deliver maximum brightness and contrast



FIRST FOCAL PLANE

FFP reticles that zooms proportionately with the magnification



CRUX ARMOR

Type-III Hard Anodization stays intact from unwanted scratch



WATERPROOF

Submersion in water without water intrusion



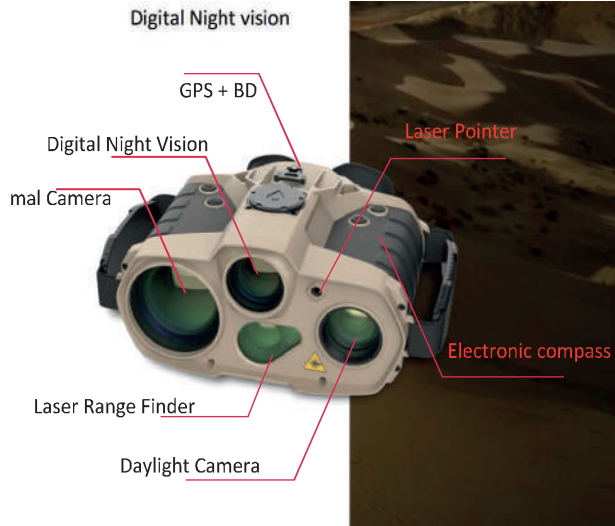
SHOCKPROOF

Defies repeated 1200G Impact

ESS100016



ESS100017



Specification

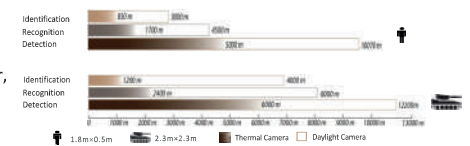
- ❖ Thermal Camera
Resolution: 640*512, 12μm
Spectral: 8~14μm
FOV: 6.1°×4.8°
- ❖ Daylight Camera
Resolution: 460M Pixel
FOV: 4.6°×3.7°
- ❖ Digital Night vision Camera
Resolution: 750×600
FOV: 6.8°×5.5°
- ❖ Laser Range Finder
Wave: 535nm
Max measure Range: 6km
(2km, 4km Optional)
Accuracy: 2m
- ❖ GPS + BD
Horizontal positioning accuracy: 5m
Vertical positioning accuracy: 10m
- ❖ Digital Compass
Azimuth measurement range: 0°~360°
Azimuth measurement accuracy: 0.5°(RMS)
Pitch angle measurement range: -90°~+90°
Pitch angle measurement accuracy: 0.4°(RMS)
Inclination angle measurement range: -180°~+180°
Inclination angle measurement accuracy: 0.5°(RMS)
- ❖ Laser Pointer
Wave: 830nm
Security Level: Class IIIA
- ❖ Display
1280×1024 OLED
- ❖ Storage
10000 JPG&4h AVI
- ❖ Diopter Range
-4~+4
- ❖ Weight
≤2.1kg (Include battery)
- ❖ Battery life
≥8h
- ❖ Dimension
198×210×105mm
- ❖ Interface
External power supply /
USB / PAL / RS232 / HDMI
- ❖ Operating Temperature
-40°C~+55°C
- ❖ Protection level
IP67



Dipper-B Multi Function Binocular

Dipper is a binocular integrated with daylight camera, thermal camera, digital night vision camera, laser range finder, laser pointer, GPS, BD, Digital compass. And all equipment fused by intelligent software and smart operation interface. It is your best device for rescue and investigation.

Working Distance

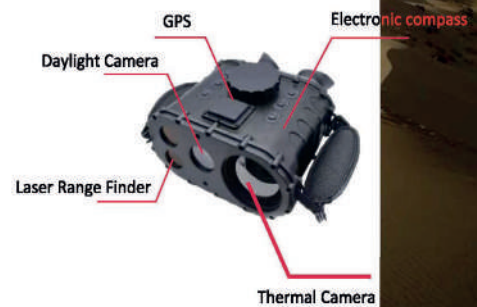


ESS100018

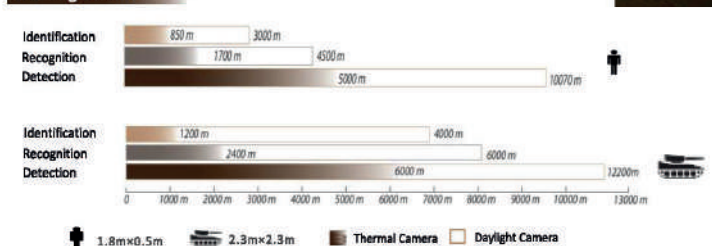
Dipper-A Multi Function Binocular

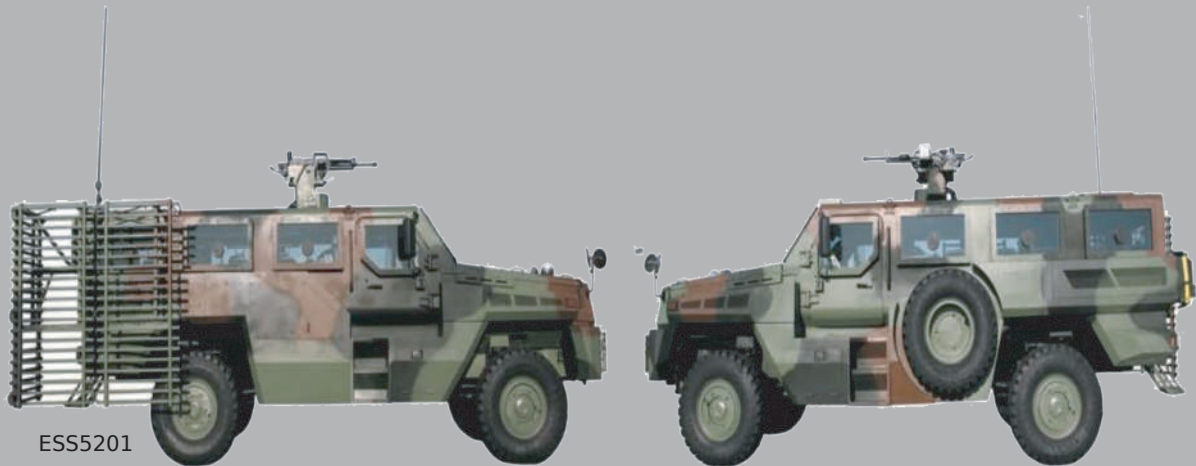
Dipper-A is a binocular integrated with daylight camera, thermal camera, laser range finder, GPS, Digital compass. And all equipment fused by intelligent software and smart operation interface. It is your best device for rescue and investigation.

- ❖ Thermal Camera
Resolution: 640*512, 12μm
Spectral: 8~14μm
Lens: 71mm, F# 1.0
FOV: 6°×4.5°
Digital Zoom: 1~6x continuous
- ❖ Daylight Camera
Resolution: 1/2.8 CMOS 1920×1080
Lens: 5.1mm~51mm 10x optical zoom
FOV: 6°×3.3°~50°×30°
Digital Zoom: 1~4x continuous
- ❖ Laser Range Finder
Wave: 905nm
Max measure Range: 2km (5km Optional)
Accuracy: ±1m
- ❖ GPS
Horizontal positioning accuracy: ≤5m
Vertical positioning accuracy: ≤10m
- ❖ Digital Compass
Accuracy: ≤1°
- ❖ Display
1920×1080 OLED
- ❖ Storage
128GB SD Card
- ❖ Weight
≤1.9kg (Include battery)
- ❖ Diopter Range
-4~+4
- ❖ Battery life
≥6h 18650x 4 pcs
- ❖ Dimension
≤210mm×196mm×100mm
- ❖ Interface
USB, Pal Video, RS232
WIFI(Optional)
- ❖ Protection level
IP67
- ❖ Operating Temperature
-40°C~+55°C



Working Distance





ESS5201

General Data

Total Combat Weight	15,000 kg
Effective Load	2,200 kg
Kerb Weight	13,080 kg
Overall Dimensions (without Turret)	6890mm x 2496mm x 2600mm
Approach/Departure Angle	32°/36°
Seating Capacity	1+1+10; 1 driver, 1 co-driver, 10 soldiers
Drive Type	4X4 Full Time
Seats	four point safety belt, anti-mine protection seat(10 units)
Tyre	4+1; YK-395/65R20; Run-flat Insert
A/C	Roof Mounted, Integral Air Conditioner; 8KW
Rear view system	7" LCD on dashboard, infra-red camera at rear end

Maneuverability

Max. Cruising Speed on Road	115 Km/h
Fuel Autonomy Range	700 Km
Fording Depth	1100 mm
Gradient	60%
Side Slope	35%
Vertical Obstacle	400 mm
Min. Ground Clearance	400 mm

Chassis

Engine	WEICHAU WP10.336, 336HP (247KW) at 2200rpm; 1,250N.m at 1,200-1,600 rpm
Transmission	CQ28 Automatic Transmission, SHAANQI (CATERPILLA technology) 6 Forwards & 1 Reverse Mechanical speeds
Steering	Hydraulic Power Steering
Brake	Pneumatic Brake with ABS
Suspension	Double Wishbone Independent Suspension/Coil Spring
Electrics	24V system; Negative ground; Battery 2 x 180Ah

Protection Capacity

Ballistic Protection	Front: Resists 12.7mm shell at 100m; Side and Rear: Resists 7.62mm explosive shell at 100m; All Sides: 86, 87, European standard EN1063
Anti-Mine Protection	Anti-mine Protection seats; Resists 8KG TNT equiv. underneath with add-on composite plate
Weapon Station Module	12.7mm MG and 35mm automatic grenade launcher
Electric winch	5T
Anti RPG	grid structure (Optional)
Central Tyre Inflation System	Available
Optional Devices with Additional Cost	VRC-2000A ultra-short wave fixed-frequency; 50W power amplifier; Truck mountable; Center-fed antenna; Intercom model ACT-1, one unit; Low light night vision device with night vision devices identify person within 100m and vehicle within 150m in the absence of moon light & star light; Command Helmet



13P-Troop Carrying
Combat Vehicle
6x6
ESS5202



YGB9402 Tank Transporter
ESS5204



15P-Heavy Duty Infantry Fire Assault
Vehicle 09ZHT4-105
ESS5203



ESS5205

12P-Mine-Resistant, Ambush-Protected (MRAP) Vehicle



ESS5206

08PG-Protected Transport Vehicle





ESS5301 APC



ESS5302 UTILITY



ESS5303 LAW ENFORCEMENT



ESS5304 CIT



ESS5305 AMBULANCE



ESS5306 WEAPON SYSTEM



ESS5307 RECONNAISSANCE

VARIANTS

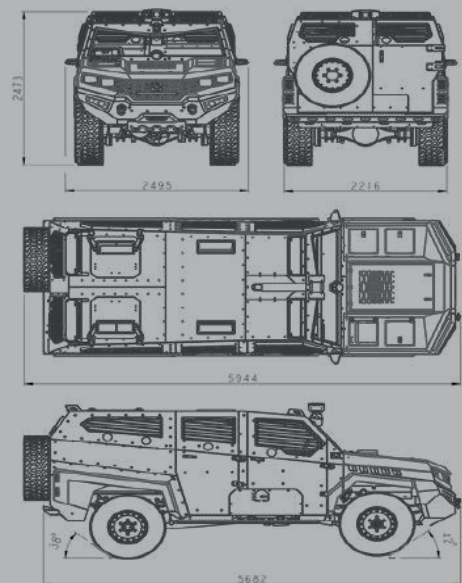
Whether it is armoured personnel carrier, law enforcement, reconnaissance, riot control, security, utility, armoured ambulance or high value asset transport, the **Ghost** can be tailored to suit your specific range of operational requirements.

TECHNICAL SPECS

Engine	Turbo Charged Intercooled Common Rail
4x2 Manual	96kW / 330Nm
4x4 Manual	110kW / 402Nm
4x2 AMT	96kW / 330Nm
4x4 Automatic	196kW / 706Nm
Brakes	ABS
Max Speed	120km/h
Low range	On all 4x4 Models
Width	2488mm
Height	2473mm
Length	5944mm
Fording Depth	1000mm
Ground Clearance	335mm Under Axle
Approach	42°
Departure	38°
Side slope	30°
Gradient	70°
Turning Circle	14m
Air Conditioner	13kW
Seating Capacity	1+11
Protection Level	B6 (Upgradeable)

MODULAR DESIGN

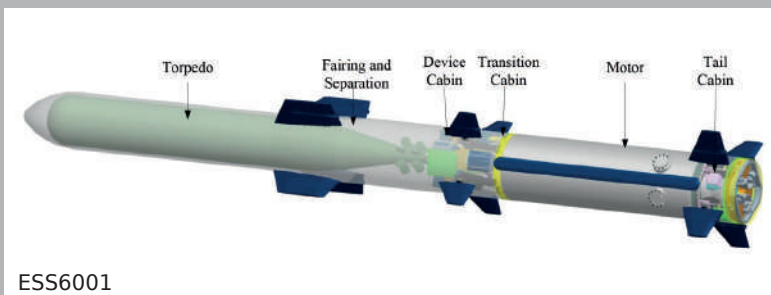
Ghost's modular design allows for easy transformation to different roles, while advanced armoured materials provide superior ballistic and blast protection in the vehicle's weight class. With the ability to carry up to 12 crew members in comfort and with ease of access through multiple entry points, the **Ghost** provides a reliable and practical solution for any operation.



LQ-008 Antisubmarine Rocket Assisted Torpedo System

System Composition

LQ-008 consists of torpedo, fairing and separation cabin, device cabin, transition cabin, motor and tail cabin. Launching by ship, the LQ-008 Antisubmarine Rocket Assisted Torpedo System, reaches predetermined in-water point with rocket assisted flight, automatically searches, tracks and attacks enemy submarines. It can be equipped in hundred-ton class missile patrol boats or 1000t-4000t class small and medium-sized destroyer-frigates. It is matched with export lightweight torpedo ET60, ET-52C and other type of combat load and often used for striking conventional power submarine and nuclear power submarine.



ESS6001

Main Features

- Long Range
- High Accuracy
- Fast Flight Speed
- Good Adaptability of Platform and Well Matching Suitability of Torpedo

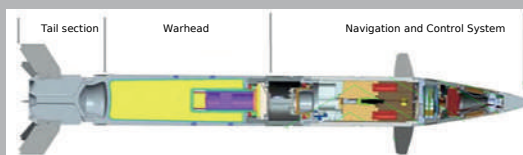
Matching Payload	China's exported lightweight torpedoes such as ET52C and ET60
Equipped Platform	patrol boat of several hundred tons or 1000t-4000t small and medium-sized frigates
Maximum Range	30 km
Minimum Range	6 km
In-water Impact Accuracy	CEP ≤ 500 m
In-water Impact Speed	50 m/s
Torpedo Weight	234.5 ~ 238 kg
Lift-off Weight	787.5 ~ 791 kg
Flight Phase Guidance	Strap down INS + Altimeter
Ammunition Load of Launch Canister	4 rounds
Launch Preparation Time	≤ 10 min
Launch Reaction Time	From taking firing order to torpedo launching ≤ 12 s
Storage Life	10 years

WS-35 Guided Projectile

WS-35 Guided Projectile is used to destroy armored vehicles, artilleries, missile position, field fortification, buildings and annihilate the enemy effectives. WS-35 Guided Projectile has the advantages of fast reaction, high accuracy and strong damage which can significantly enhance the artillery's strike ability.



ESS6002



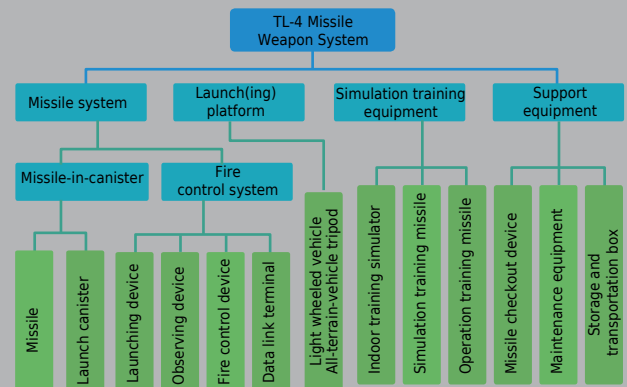
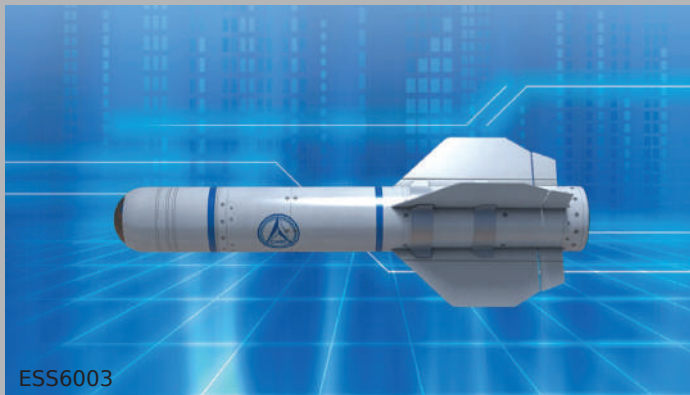
Main Features

- Fast Reaction
- High Accuracy

Caliber	155 mm
Weight	50 kg
Length	910 mm
Range	18~40 km
Guidance Mode	GPS/BeiDou Navigation Satellite System+Inertial guidance
Type of Warhead	Blast-Fragmentation /Fragmentation-Explosion-Inflammation /FAE
Detonation Mode	Proximity Fuze/Impact Fuze/Delayed Action Fuze/Time Fuze
Accuracy	CEP ≤ 10 m
Intensive Kill Radius	> 26 m
Reliability	0.9
Storage Life	10 years

TL-4 Missile Weapon System

TL-4 missile adopts image-homing + duplex wireless data link navigation system, so it is capable of two different operational modes which are fire & forget and fire & update. TL-4 can be equipped with TV seeker or un-cooled infrared seeker. Two different attack modes including direct-attacking and top-attacking trajectory could be selected according to the target type. TL-4 missile can be loaded on multi-platforms, including light vehicle, armed helicopter, UAV, ATV or a portable tripod to perform striking on high value targets including enemy tanks, armored vehicles, fortifications and fire point etc.



Main Features

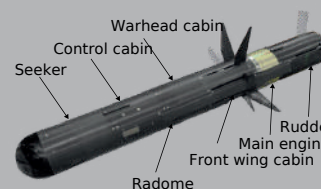
- Simple Operation
- Multi-platform Capability
- Day and Night Combat
- Strong Mobility and Survivability
- Strong Battlefield Environment Adaptability

Firing Range	0.5~6 km
Guidance Mode	Uncooled infrared/TV image homing + duplex wireless data link
Kill Probability	≥ 85%
Missile Mass	≤ 26 kg
Missile Length	≤ 1,200 mm
Missile Diameter	140 mm
Temperature	-40°C ~ +70°C
Altitude	≥ 4,500 m
Operation Mode	Lock on before launch, Fire and forget; Lock On after Launch, Man in Loop, Aiming Correction

TS-01 Light Multi-purpose Missile Weapon System

TS-01 is a shoulder-fired infrared imagery guidance multipurpose weapon system with quick response and accuracy in attacks. It can be used both individually and in groups. Equipped with uncooled infrared imaging guidance, the missile lock on the target before launch and track the target automatically after launch, which made it flexible in use. The weapon system is mainly used for precise attack to enemy's main tanks, armored vehicles, vehicles, fortifications, firing point and effective strengths.

ESS6004



Main Features

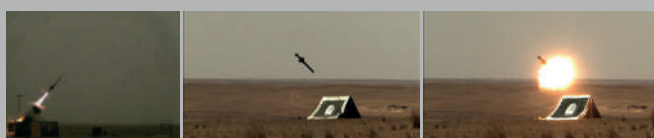
- Quick Response
- High Precision
- Lock on Before Launch
- Fire-and-forget
- High Survival Rate

Weight of Missile-in-container	≤ 15.5 kg
Length of Missile	1053 mm
Diameter of Missile	127 mm
Effective Range	200-3,000 m
Armor-penetration Depth	≥ 800 mm (homogeneous steel plate)
Average Hit Probability	≥ 85%
Operating Range of Launch-control	The infrared range: ≥ 3,500 m The visible light range: ≥ 4,000 m
Guidance System	Uncooled infrared imaging homing guidance
Launching Platform	Individual action
Attack Mode	Top attack and direct attack
Type of Warhead	Tandem shaped-charge fragmentation warhead Armor-penetration fragmentation warhead

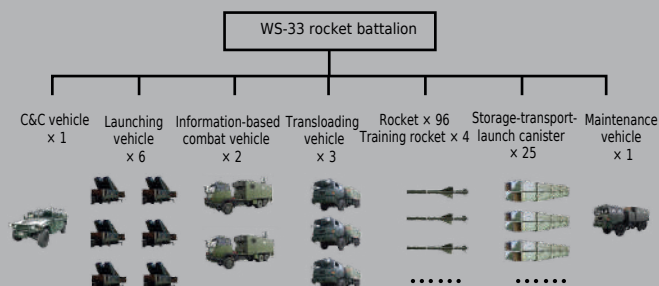


WS-33 Precision Guided Multiple Launch Rocket System

WS-33 precision GMLRS adopts whole course guidance technology on the basis of combined navigation and infrared imaging terminal guidance technology. During flight, the rocket connects with tactical network via data link to acquire the information of a moving target. The data link resets the position and contrastive images of the target. After the target is captured and locked by the seeker, WS-33 will actualize precision attack against the target.



Range	20-60 km
Accuracy	CEP ≤ 10 m
Take-off Mass	240 kg
Length of Rocket	3,770 mm
Diameter of Rocket	φ200 mm
Guidance Mode	SINS / GNSS + IR imaging terminal homing guidance
Launching Readiness Time	≤ 7 min
Weight of Warhead	30 kg
Type of Warhead	Blast-fragmentation warhead
Loading Quantity	8 per vehicle
Launch Mode	Vehicular canister-type oblique launch

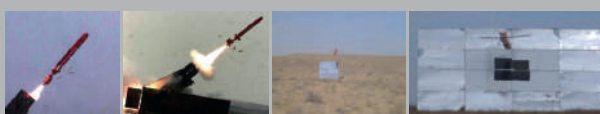
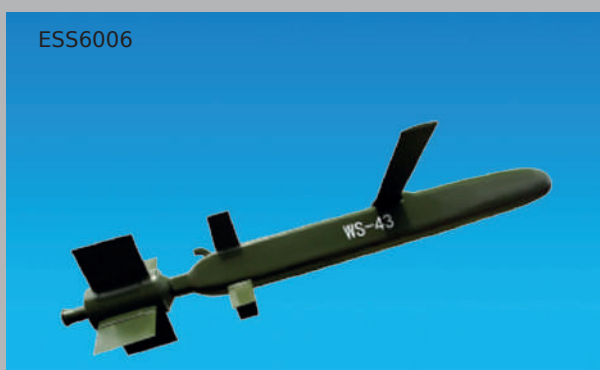


Main Features

- High Accuracy
- High Vehicle Maneuverability
- Quick Combat Reaction
- Easy Operation and Simple Maintenance

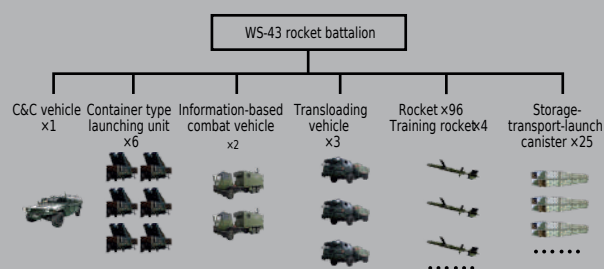
WS-43 Loitering Attack Weapon System

WS-43 loitering rocket is a new concept weapon ammunition. The loitering rocket utilizes networked control system and data link system to achieve netting operation, online track planning and other armament coordinated operations. With integrated operation capability of target reconnaissance, control, striking and assessment, it can effectively cope with stationary target, moving target, reverse slope target and time-sensitive target, etc.



Main Features

- Multi-Function: Investigation, Blockade and Attack
- Transparent on Battlefield and Cooperative on Tactics
- Quick Combat Reaction
- Selective Attack on Mobile Target
- Multiple Combat Mode and Agile Combat Operation



Operational Radius	20-60 km
Endurance	≥ 30 min
Loitering Speed	Mach 0.3
Accuracy	CEP ≤ 10 m (Terminal homing guidance)
Take-off Mass	220 kg
Length of Rocket	3,421 mm
Wingspan of Rocket	2,100 mm
Guidance Mode	SINS / GNSS + Terminal homing guidance
Weight of Warhead	20 kg
Type of Warhead	Blast-fragmentation warhead
Loading Quantity	8 per vehicle
Launch Mode	Canister-type oblique launch



ESS7004





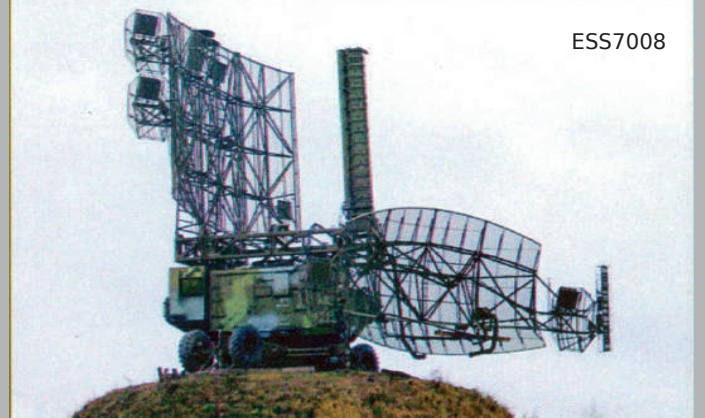
ESS7005



ESS7006



ESS7007



ESS7008



ESS7009



ESS7010

ESS7011



ESS7012



ESS7013



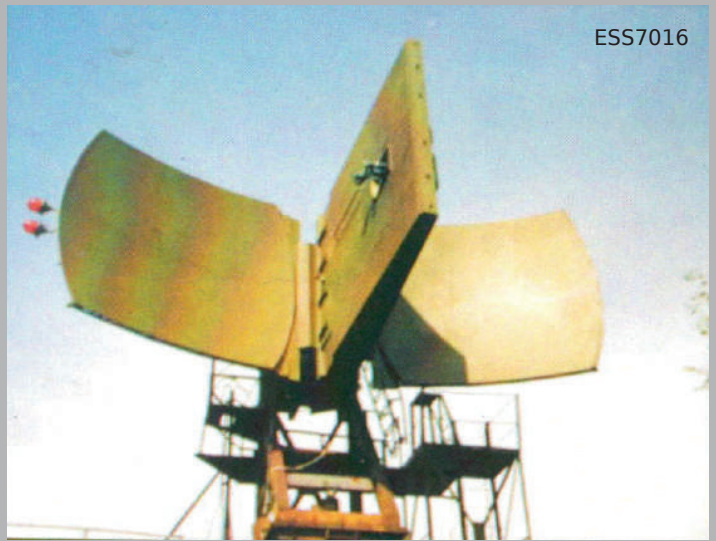
ESS7014



ESS7015



ESS7016



Underground Metal Detector Model UMD-II



Description

The UMD-II is a versatile multi-purpose metal detector suitable for police, military and civilian users. It addresses requirements for crime scene and area searching, explosive ordnance clearance. It is approved and used by police services around the world. The new detector introduces simplified controls, an improved ergonomic design and advanced battery management. It is weather resistant and designed to withstand extended periods of use in harsh environments while provide a high level of sensitivity.

For operator confidence, a status LED illuminates green when the unit is on and operating correctly. Target detection is indicated by an intuitive LED array and audio tone, delivered by an internal sounder or optional earpiece.

The instrument is powered with three rechargeable 'D' cells, which can continuously work for 12 hours.

The UMD-II includes easily interchangeable detection heads: a robust halo for rapid area search, a probe for searching drains, culverts, hedges and undergrowth. The electronics are assembled and tested by computer controlled equipment for a high degree of reliability and are housed in a slim, rugged and ergonomic case.

Key Features

- ▶ Target detection indicated via LED display and audio tone.
- ▶ Three preset sensitivity levels.
- ▶ Interchangeable detection heads: Halo for rapid area search, Probe for drains & culverts.
- ▶ Automatic self-test and calibration for operator confidence and ease of use. Optimized power consumption.
- ▶ Low battery indication.

Specification

Electronic Technique	Single 2.4mm PEC double surface mounted technology, the processor is based on 8-bit 2*RISC ADC (8-bit 2*instruction set AD converter)
Battery	3 LEE LR20 Manganese Alkaline Dry cell
Battery life	10-18 hours
Packing case	ABS case
Device Weight	Halo 2.1 Kg; Probe 1.65 Kg
Gross Weight	12Kg (device+case)
The detecting pole length	Halo: 1080mm~1370mm; Probe: 1135mm~1395mm
Operating and Storage temperature	-25°C~60°C

Item No.	Target Size	Detection range at low level	Detection range at medium level	Target Picture
1	268x74x144mm	30cm	40cm	
2	298x78x186mm	25cm	36cm	
3	307x54x184mm	16cm	32cm	
4	347x82x195mm	25cm	33cm	
5	275x62x134mm	17cm	32cm	
6	Coin, D25mm 6gram	7cm	16cm	

ESS8001

Mine Detector Model UMD-III



Description

UMD-III mine detector is a widely used hand-held (single-soldier operating) mine detector. It adopts high frequency pulse induction technology and it is highly sensitive, especially suitable for detecting minor metal mines. The operation is simple, so the operators can use the device only after a short training.

Features

1. Waterproof, which can be detected under water.
2. Being controlled by a microprocessor with accurate timing, fast conversion and strong signal processing capability.
3. Super sensitivity to identify very small metal objects.

Technical Parameters

Weight	2.1kg
Transport weight	11 kg (device+case)
The detecting pole length	1100m~1370mm
Battery	3LEE LR20 Manganese Alkaline dry cell
Battery life	At maximum sensitivity - 12 hours At medium and low sensitivity - 18 hours Low voltage alarming by sound and light
Operating humidity	Fully enclosed and be able to operate 2 m under water.
Operating temperature	-25°C~60°C
Storage temperature	-25°C~60°C
The detecting coil	The longest detecting pole is 965mm, the shortest is 695mm, and weight 1300g. Glass resin telescopic rod, the surface is coated to protect the environment. The size of detecting coil is 273mm*200mm, black ABS material, surface is treated with EMC, and a hybrid RX coil is used to improve the signal/noise ratio.

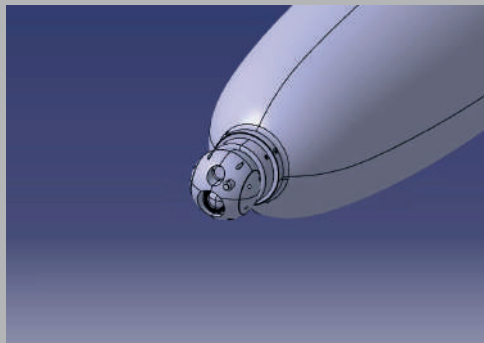
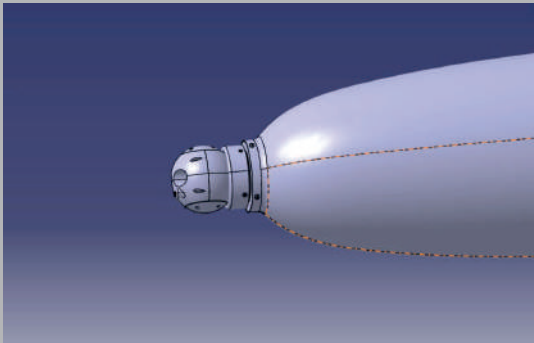
Item No.	Target Size	Detection range at low level	Detection range at medium level	Target Picture
1	268x74x144mm	28cm	38cm	
2	298x78x186mm	23cm	34cm	
3	307x54x184mm	14cm	30cm	
4	347x82x195mm	23cm	31cm	
5	275x62x134mm	15cm	30cm	
6	Coin, D25mm 6gram	6cm	15cm	

ESS8002

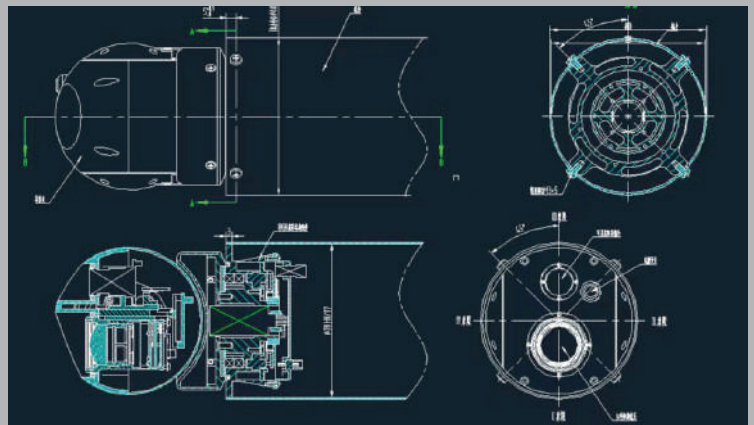




ESS9001



Bomb Load/Maximum Range	40KG
Speed	1500KM
Flying time	160-220km/h
MTOW	10h
Flying control system	120kg
Avionics system	1kg
Fuel weight	80L
Dimension Length	2.85m
Wing	2.4m
Wing area	3.5m2





HC-342H

✕	DIAGONAL DISTANCE	1960mm
⌚	TAKE-OFF WEIGHT	51kg
⬇	PAYLOAD	15kg
🕒	MAX ENDURANCE	120min
🕒	CRUISING SPEED	0-15m/s
📶	CEILING	1600m

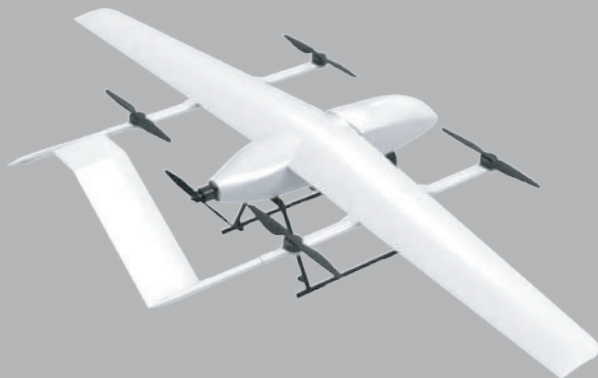
ESS9002



HC-332E

✕	DIAGONAL DISTANCE	1560mm
⌚	TAKE-OFF WEIGHT	19kg
⬇	PAYLOAD	5kg
🕒	MAX ENDURANCE	60min@2kg
🕒	CRUISING SPEED	0-15m/s
📶	CEILING	3000m

ESS9003



HC-525E

✕	WINGSPAN	3200mm
⌚	TAKE-OFF WEIGHT	22kg
⬇	PAYLOAD	3kg
🕒	MAX ENDURANCE	2h
🕒	CRUISING SPEED	80- 90km/h
📶	CEILING	3000m

ESS9004

ESS12001

BARUT

PRODUCT Gunpowder for sporting & hunting ammo 7.62x39
 PARTY № TP-6 29/2022T
 PARTY QUANTITY, kg 4.948
 YOM 2022
 QUALITY REQUIREMENTS TDS 7277-180-07506104-2011 mod. 7

**PHYSICAL AND CHEMICAL CHARACTERISTICS**

	TDS ref.	Tests Value	Value after thermal aging
volatile substances, %			
non-removable	> 0,2	0,400	
removable	0,6-1,6	0,900	
total	< 2,9	1,300	
mass fraction of additives, %			
Diphenylamine	1,0-2,0	1,500	
graphite	< 0,4	0,390	
camphor	< 1,8	1,200	
bulk density %	NA	0,862	
geometric dimensions, mm.			
the thickness of the burning vault	0,18-0,25	0,210	
diameter of the channel	0,07-0,15	0,120	
length	0,85-1,25	0,960	
resistance measured by manometric method, mmHg.	<220	69	

BALLISTIC TESTS RESULTS

Test goals	Selection of charge weight
Test place	Tambov gunpowder factory
Test system	7.62 ballistic barrel chambered in 7.62 x 39 caliber
Exemplary bullet	7.62x39 124 GR. FMJ
Exemplary case	7.62x39 Steel
Test selection of product	Average samples

	TDS ref.	Tests value	Exemplary shots value
Charge weight, grams	1,50-1,62	1,57	
Testing shots quantity	10	10	10
Initial speed V_{25} , m/s			
average	720 ± 5	719	708
max		729	715
min		712	701
Speed dispersion ΔV_{25} , m/s	≤ 25	17	14

Test goals	Determination of powder gases pressure, kgf/cm ²
Test place	Tambov gunpowder factory
Test temperature	+20 °C
Exemplary bullet	7.62x39 124 GR. FMJ
Exemplary case	7.62x39 Steel
Test system	Average charges with weight selected

	TDS ref.	Tests value	Exemplary shots value
Charge weight, grams		1,57	
Testing shots quantity	10	10	10
Powder gases pressure, kgf/cm ²			
average	≤ 3000	2429	2409
max	≤ 3400	2612	2496
min		2335	2350



ESS12002

NITROPOTASYUM

**POTASSIUM NITRATE - PYROTECHNIC
CHEMICAL FOR DEFENCE USE
CAS NO: 7757-79-1
UN NO: 1486**



PARAMETERS	TEST RESULTS	LIMITS
Physical Form	Coystaline Form	SIZE 01 MM- 08 MM
MOISTURE CONTENT	0.32 %	MAX 1.0 %
pH LEVEL (10% SOLUTION)	6.80	6.0 TO 8.0
CHLORIDE (CL)	LESS THAN 0.02.%	MAX 0.05 %
WATER SOLUBLE	LESS THAN 400 PPM	MAX 800 PPM
PURITY	99.62 %	MIN 99,0 %
NITROGEN Content	13.71 %	13.0 % - 13.8 %
POTASSIUM OXIDE (K2O)	46.04%	MIN 46.0 %
COLOUR	TRANSPARENT WHITE	TRANSPARENT WHITE
ODOUR	ODOURLESS	ODOURLESS

PARAMETERS	TEST RESULTS	LIMITS
Physical Form	Powder Form	100 % PASSING IN 30 MESH
MOISTURE CONTENT	0.28% MAX 1.0 %	0.28 % MAX 1.0 %
pH LEVEL (10% SOLUTION)	6.90 6.0 TO 8.0	6.90 6.0 TO 8.0
CHLORIDE (CL)	LESS THAN 0.02 % MAX 0.05 %	LESS THAN 0.02% MAX 0.05 %
WATER SOLUBLE	LESS THAN 0.01 % MAX 0.02 %	LESS THAN 0.01 % MAX 0.02 %
PURITY	99.62 %	MIN 99.0 %
NITROGEN Content	13.71 %	13.0 % - 14.00 %
POTASSIUM OXIDE (K2O)	47.12%	MIN 46.0 %
COLOUR	TRANSPARENT WHITE	TRANSPARENT WHITE
ODOUR	ODOURLESS	ODOURLESS
CA & MG	LESS THAN 0.03 %	MIN 0.05%
SULPHATE	LESS THAN 0.015 %	MAX 0.03 %
HEAVY METALS(AS,HG,CD,PB,CR)	LESS THAN 0.001 %	MAX 0.002 %



ESS12003

TNT

Appearance: Scale like

Color: Light yellow to yellow Solidification point: $\geq 80.20^{\circ}\text{C}$ Moisture and volatile matters: $\leq 0.07\%$ Acidity (as H_2SO_4): $\leq 0.010\%$ Insoluble in acetone (benzene or toluene): $\leq 0.05\%$

25kg per bag



ESS12004

RDX / BINOCULARS

ATTRIBUTE	SPECIFICATION	TEST RESULTS
Acidity (in terms of HNO_3):	0,05%	0,03%
Insoluble in Acentone	0,05%	0,03%
Insoluble particals retained on #60	5 particles max	0
Melting point	200°C	200°C
Granulation% Passing		
Mesh#20	98+2%	100%
Mesh#50	98+10%	96,4%
Mesh#100	60+30%	39,2%
Mesh#200	25+20%	5,6%
Appearance under the microscope	Crystalline agglomerates free of needle shapes	Compliant

DEACRIPTION	UNIT		
Melting point	c	>200	203,8
Acidity (as HNO_3)	Wt%	50,05	50,05
Insoluble particles remaining on US standard sieve #50	particies	55	0
Total acetone insolubel material	Wt%	50,03	0,015
Total acetone insolubel material	Wt%	50,03	0,004
Particio size distribution			
Through US standard sieve #20	%	95+100	100
Through US standard#50	%	70+100	94,3
Through US standard#50	%	20+90	69,0
Impact sensitivity by kast test	%	584	44

ESS12005

NITROSELULOZ**SPECIFICATION NITROCELLULOSE**

INDEX	NORM
1.Acetone insolubl,%not more	0.4
2. Ash,%not more	0.4
3. ether/alcohol Solubility ,%	28-40
4.Finenss,mL	70~110
5.Moisture,% not more	27~35
6.Bergmen junk stability test, ml NO/g,	2.5
7.Nitrogen,%	13.10~13.2
8.Nitrogen Content-Grade C Type 1 (Preferred Value: 13.15)	0
9. Viscosity,mPa s(cP,0E),Seconds	13.10~13.2
	0
	9~15





**EMN DEFENCE
SANAYİ ve DIŞ TİCARET A.Ş.**

Hürriyet Bulvarı No:129 / 4 Beylikdüzü/ İstanbul / Türkiye
Tel.: 0212 999 05 69 e-mal: info@emn defence.com